

Impact of Body Mass Index on Radiation Doses in Interventional Radiology

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1. INTRODUCTION

Interventional procedures are essential but expose patients to high doses of radiation, especially those who are overweight or obese [1]. Since traditional dose indicators are based on “standard” patients, they become less representative of the current population [1,2]. This study aims to propose Typical Values (TVs) stratified by Body Mass Index (BMI) and also compares different procedure types, providing more accurate indicators of clinical practice.

2. MATERIALS AND METHODS

• **Equipment:** GE Innova IGS 320 (GE Healthcare)

• **Procedures:** Vascular Angiography (25), Vascular Angioplasty (68), Coronary Angiography (1590), and Coronary Angioplasty (517)

• **Sample:** 2,200 retrospective examinations (Dec/2022 – Mar/2024), patients ≥ 18 years with collected data on:

- Weight, height and reference point air kerma ($K_{a,r}$ in mGy)

• **Determination of TVs:**

- TVs were defined according to ICRP 135 [3] → **median** as the reference statistic.

Three levels of analysis:

- **Overall:** all examinations combined → institutional TV.
- **Procedure:** Differentiation among collected procedures.
- **BMI:** According to the World Health Organization (WHO): underweight (BMI < 18.5 kg/m²), normal weight (18.5–24.9 kg/m²), overweight (25.0–29.9 kg/m²), obesity class I (30.0–34.9 kg/m²), obesity class II (35.0–39.9 kg/m²), and obesity class III (≥ 40.0 kg/m²) [4].

3. RESULTS



TV $K_{a,r}$ global: 413,0 mGy

Figure 1 shows boxplots of the $K_{a,r}$ distribution for each type of procedure.

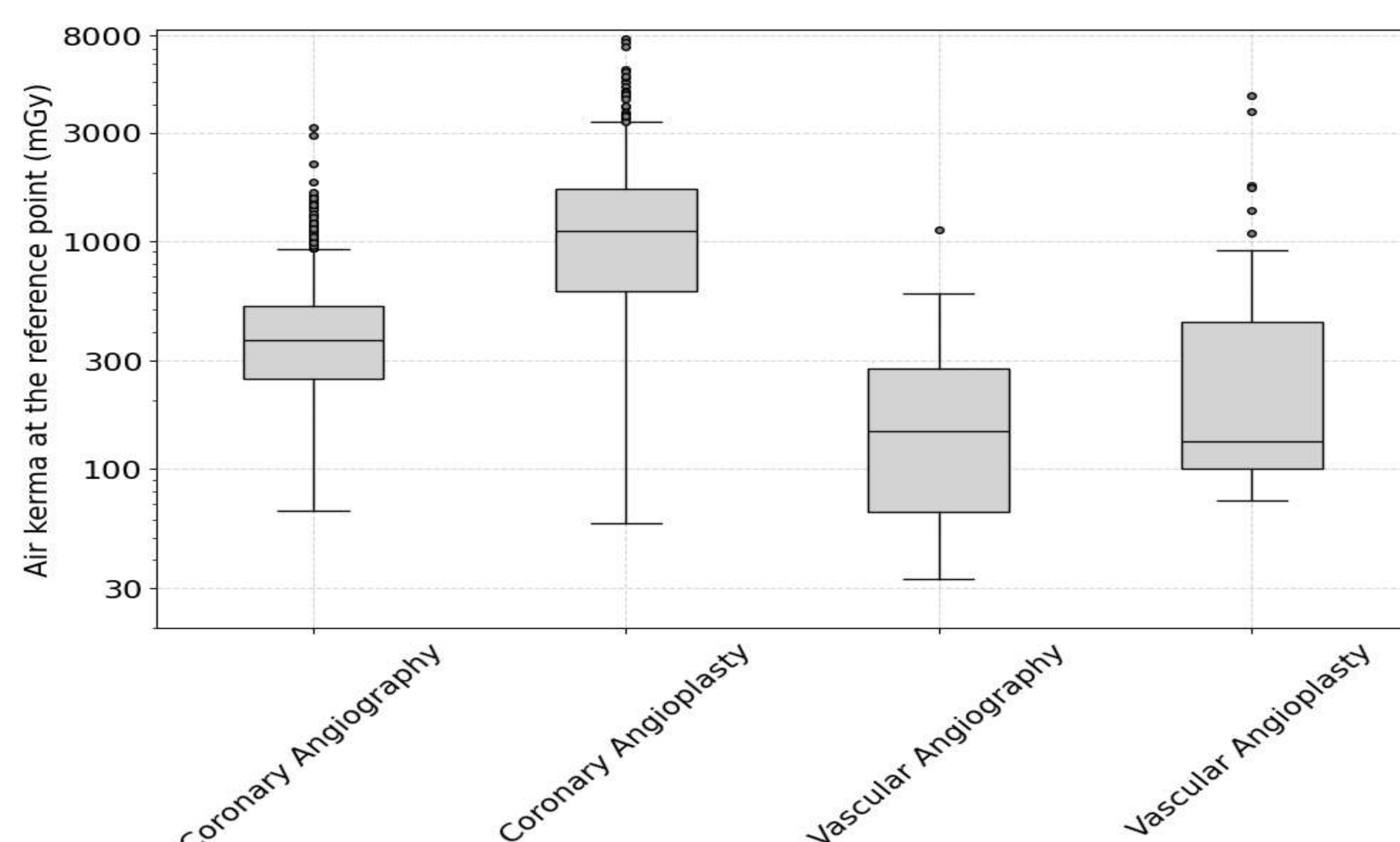


Figure 1: Distribution of $K_{a,r}$ values by type of interventional procedure.

Figure 2 shows the dose increasing with BMI, from 185 mGy (underweight) up to 796 mGy (obesity class III).

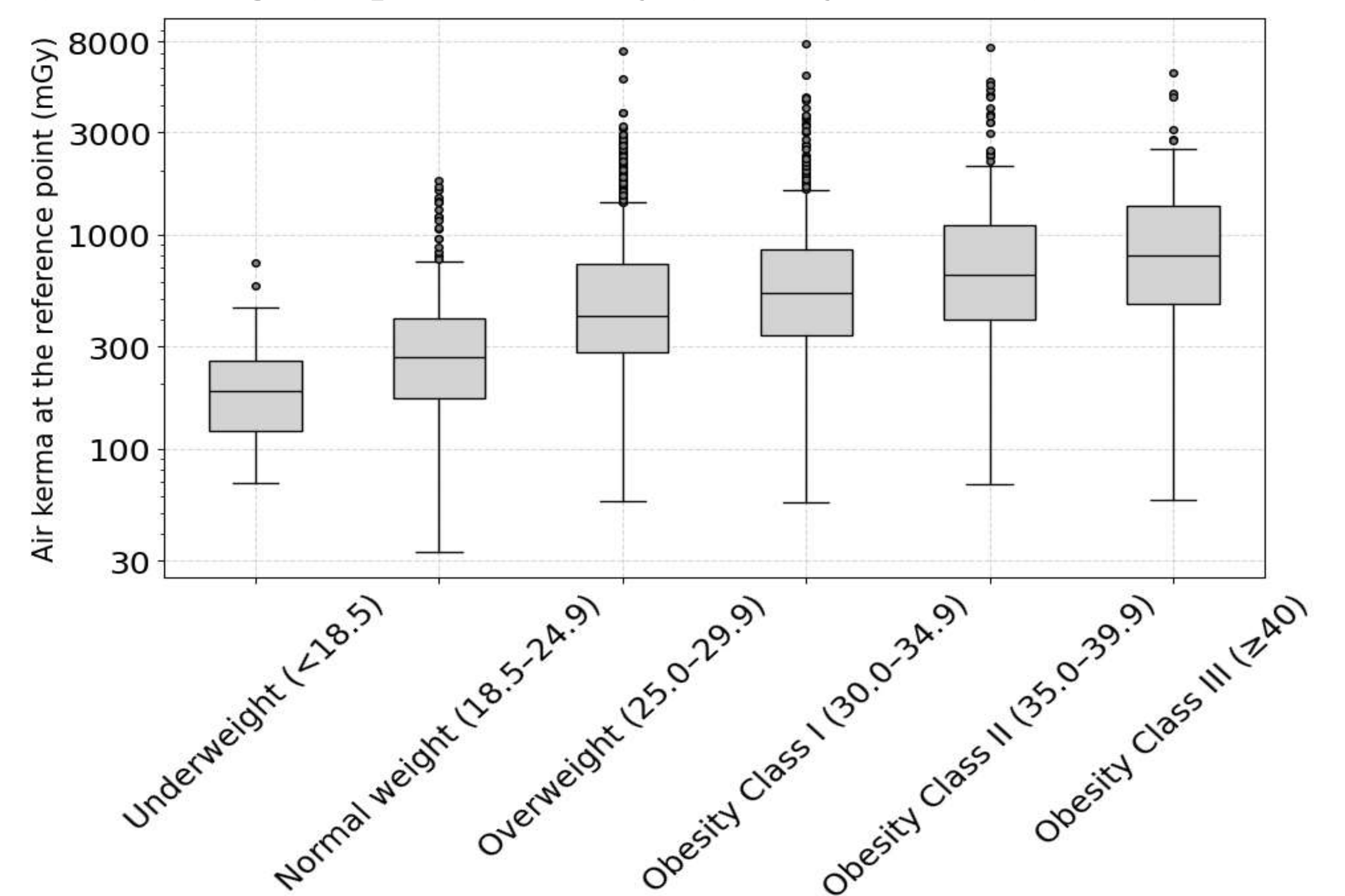


Figure 2: Distribution of $K_{a,r}$ values (mGy) by BMI category.

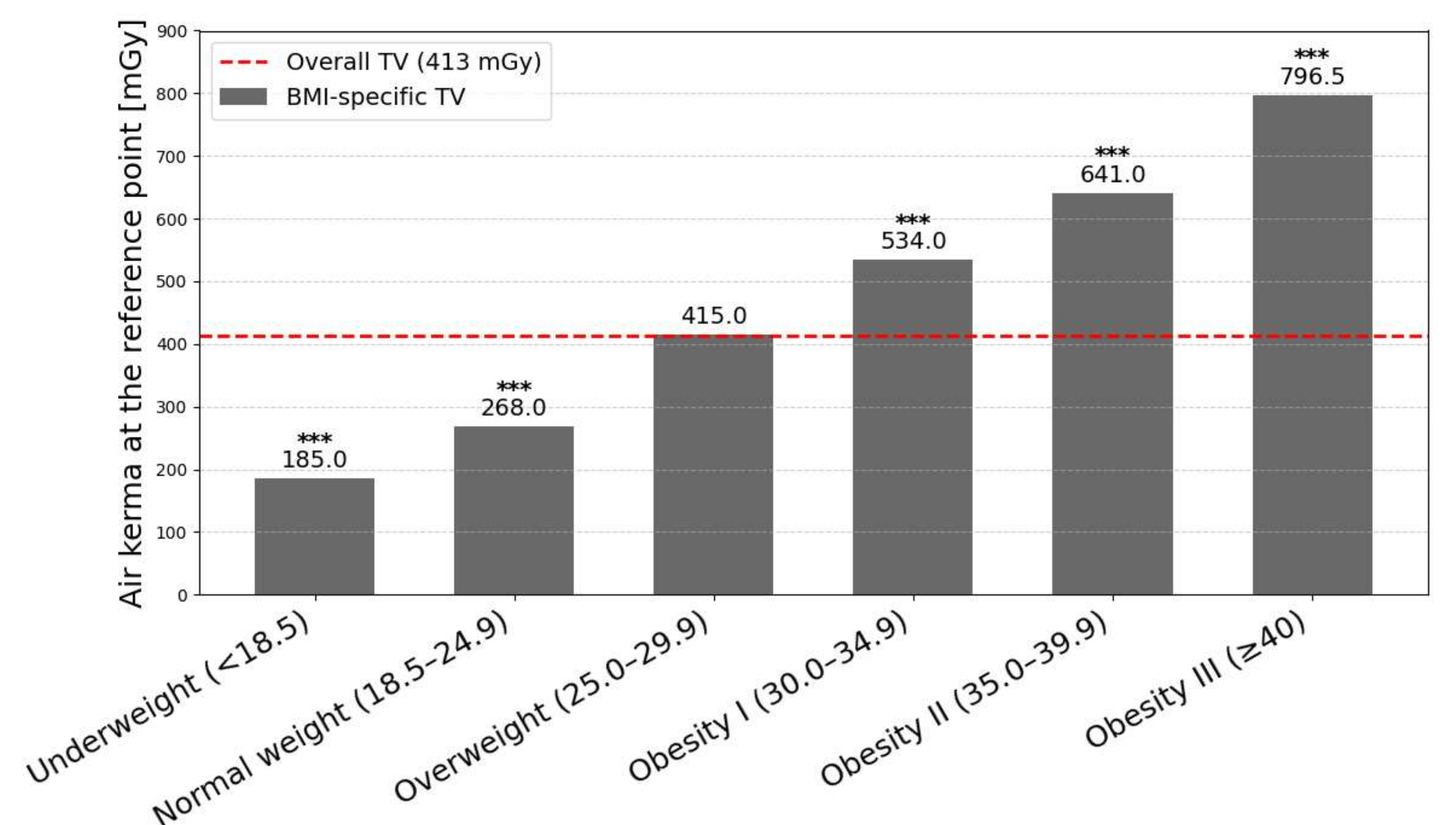


Figure 3. TVs of $K_{a,r}$ by BMI category compared to the overall TV.

4. CONCLUSIONS

The characterization of radiation exposure by procedure type and anthropometric profile provides a detailed understanding of clinical practice and contributes to a consistent assessment of patient dose, taking into account the real variability of the population served. This approach, in accordance with ICRP 135, is applicable to routine clinical practice and can be replicated in other centers.

5. REFERENCES

- [1] L. A. Vieira, et al., ‘Diagnostic reference levels based on patient body mass index for select interventional procedures in Minas Gerais/Brazil’, *Radiat. Prot. Dosimetry*, vol. 198, no. 7, pp. 379–385, Apr. 2022, doi: 10.1093/rpd/ncac048.
- [2] J. Tristram et al., ‘Typical doses and typical values for fluoroscopic diagnostic and interventional procedures’, *J. Radiol. Prot.*, vol. 42, no. 2, p. 021510, Mar. 2022, doi: 10.1088/1361-6498/ac5294.
- [3] E. Vañó et al., ‘ICRP Publication 135: Diagnostic Reference Levels in Medical Imaging’, *Ann. ICRP*, vol. 46, no. 1, pp. 1–144, Oct. 2017, doi: 10.1177/0146645317717209.
- [4] Obesity: preventing and managing the global epidemic. Report of a WHO consultation’, *World Health Organ. Tech. Rep. Ser.*, vol. 894, pp. i–xii, 1–253, 2000

